GeoVisionX

Real-Time Earth Intelligence from Orbit

The customer problem

1. Who is the customer?
→ Satellite operators, Disaster Management Agencies and Precision Agriculture Platforms

2. What's going wrong?
→ Satellites generate multi-spectral, high-volume data continuously
Downlink bandwidth is limited → transmission bottleneck
→ 90% of raw data has low informational value
The challenge is not capturing data — it's extracting value in time.

3. Why is this a problem?
→ Satellite imagery must be sent to Earth before analysis → causes delays and high bandwidth

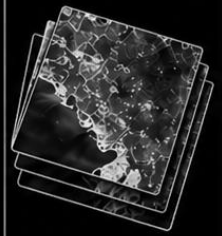
THE MODEL

TERRAMIND-POWERED MULTI-TASK EDGE AI

ONE MODEL. THREE INSIGHTS. REAL-TIME ON ORBIT.

1 INPUT

Sentinel-2
RGB Imagery



10m Resolution

RGB Bands

2 CORE

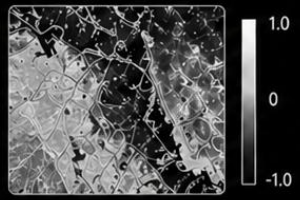
TERRAMIND-SMALL ENCODER

IBM TerraMind Small

Lightweight • Fast • Edge-Optimized
Runs in real-time on Jetson (Edge)

3 TiM REASONING LAYER

TiM (Thinking-in-Modalities)
Synthetic NDVI Generation



Infers NDVI-like representation from RGB to enable spectral reasoning and better decisions

4 MULTI-TASK HEADS

Shared Features

Flood Detection

Crop Health

Change Detection

Single encoder. Three task-specific heads. One inference pass.

5 SELF-VALIDATION SCORER

TerraMind Scorer Head
Cross-checks predictions and assigns confidence

High Confidence Only

OUTPUT

Transmit only high-confidence alerts & maps

Save Bandwidth.
Increase Trust.

FOUNDATION MODEL
Built on IBM TerraMind-Small, fine-tuned for high-speed on-orbit inference.

TiM (THINKING-IN-MODALITIES)
Model infers NDVI-like features from standard RGB imagery — enabling vegetation and water analysis without multispectral sensors.

MULTI-TASK EFFICIENCY
One shared encoder with three specialized heads delivers flood, crop, and change insights in a single forward pass.

SELF-VALIDATION SCORER
Secondary TerraMind head validates outputs and filters noise—only high-confidence alerts are downlinked.

EDGE AI FOR EARTH OBSERVATION — SMARTER. FASTER. LIGHTER.

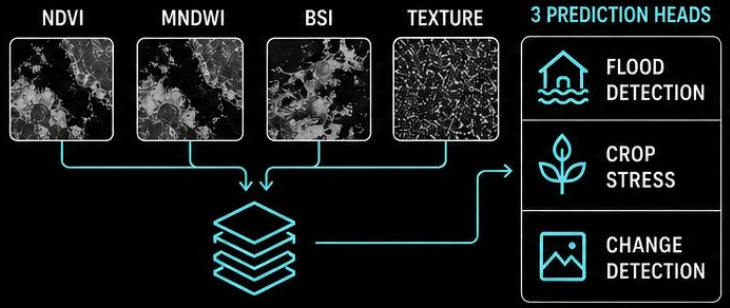
GEOVISIONX

SYSTEM SPECS & PERFORMANCE

Multi-modal, multi-task, self-validating edge AI for Earth observation.

ONE MODEL. THREE INSIGHTS.
REAL-TIME ON ORBIT.

CORE ARCHITECTURE



Fuses 4 modalities across 3 heads in a single forward pass.

BENCHMARK VS BASELINE

METRIC	BASELINE (NDVI THRESHOLD)	GEOVISIONX	IMPROVEMENT
Flood Detection (F1 Score)	0.55	0.87	+58%
Crop Detection (mIoU)	0.46	0.74	+61%
Change Detection (mAP)	0.56	0.81	+45%

CONFIDENCE & RELIABILITY

Self-Validation Scorer adds confidence to every prediction.

60 – 93 / 100

Baseline: ✕ No confidence scoring

PERFORMANCE & EFFICIENCY

- Inference: < 50 ms (CPU)
- Baseline: < 5 ms (CPU)
- Pure NumPy (No PyTorch dependency)
- Peak RAM: < 100 MB

DEPLOYMENT SPECS

Model size target: < 200 MB

- Reference: TerraMind-Small (~120 MB, INT8)
- Optimized for NVIDIA Jetson Orin NX

BANDWIDTH OPTIMIZATION

RAW SATELLITE IMAGE
~750 KB

STRUCTURED JSON OUTPUT
< 2 KB

> 99% BANDWIDTH SAVINGS

KEY TAKEAWAY

- ✓ Significantly higher accuracy across all tasks
- ✓ Self-validating with confidence scores
- ✓ Ultra-efficient edge performance
- ✓ Minimal bandwidth usage for real-world deployment

SMARTER INSIGHTS. FASTER DECISIONS. LIGHTER FOOTPRINT.
EDGE AI FOR EARTH OBSERVATION.

Honest Limits

1. **Synthetic Data Only**
   Uses demo/generated images, not real satellite data → limits real-world accuracy.
2. **Noisy Predictions**
   Adds random noise → same input can give different outputs (not reliable).
3. **RGB Only Input**
   No true multi-spectral data (like NIR) → reduces analysis accuracy.
4. **No Trained ML Model**
   Uses rule-based logic instead of a trained model → no learning or generalization.
5. **No Security / Authentication**
   API is open with no auth → vulnerable to misuse.

Improvement Plan

1. **Data & Evaluation**: Replace hardcoded metrics with real evaluation using datasets like EuroSAT
2. **Model Reliability**: Remove stochastic noise to ensure consistent and reproducible predictions
3. **Model Upgrade**: Integrate a lightweight trained model (e.g., ResNet-18) for real learning capability
4. **Performance Accuracy**: Correct bandwidth calculations using realistic compressed image sizes
5. **Real-World Data**: Replace synthetic inputs with real Sentinel-2 satellite imagery
6. **System Optimization**: Clean dependencies, fix CORS, and improve deployment configuration